

FORM TP 2016276



TEST CODE **02238020**

MAY/JUNE 2016

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PHYSICS

UNIT 2 – Paper 02

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions in TWO sections. Answer ALL questions.
2. Write your answers in the spaces provided in this booklet.
3. Do NOT write in the margins.
4. Where appropriate, ALL WORKING MUST BE SHOWN in this booklet.
5. You may use a silent, non-programmable calculator to answer questions, but you should note that the use of an inappropriate number of figures in answers will be penalized.
6. If you need to rewrite any answer and there is not enough space to do so on the original page, you must use the extra lined page(s) provided at the back of this booklet. **Remember to draw a line through your original answer.**
7. **If you use the extra page(s) you MUST write the question number clearly in the box provided at the top of the extra page(s) and, where relevant, include the question part beside the answer.**

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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LIST OF PHYSICAL CONSTANTS

Speed of light in free space	c	=	$3.00 \times 10^8 \text{ m s}^{-1}$
Permeability of free space	μ_0	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
Permittivity of free space	ϵ_0	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
	$\frac{1}{4\pi \epsilon_0}$	=	$9.0 \times 10^9 \text{ m F}^{-1}$
Elementary charge	e	=	$1.60 \times 10^{-19} \text{ C}$
Planck's constant	h	=	$6.63 \times 10^{-34} \text{ J s}$
Unified atomic mass constant	u	=	$1.66 \times 10^{-27} \text{ kg}$
Energy equivalence	$1 u$	=	931 MeV/c^2
Rest mass of electron	m_e	=	$9.11 \times 10^{-31} \text{ kg}$
Rest mass of proton	m_p	=	$1.67 \times 10^{-27} \text{ kg}$
Acceleration due to gravity	g	=	9.81 m s^{-2}
1 Atmosphere	atm	=	$1.00 \times 10^5 \text{ N m}^{-2}$
Avogadro's constant	N_A	=	$6.02 \times 10^{23} \text{ per mole}$



SECTION A

Answer ALL questions.

Write your answers in the spaces provided.

1. (a) (i) Explain how terminal potential difference, internal resistance and ideal source are related in a practical source of electromotive force (e.m.f.) such as an alkaline battery.

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[3 marks]

- (ii) State TWO qualitative practical observations which demonstrate the existence of internal resistance.

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[2 marks]

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- (b) In an experiment to determine the internal resistance of a 9 V alkaline battery, a precision resistance box is connected to the battery through a digital multimeter (DMM) set up as an ammeter. The circuit is shown in Figure 1.

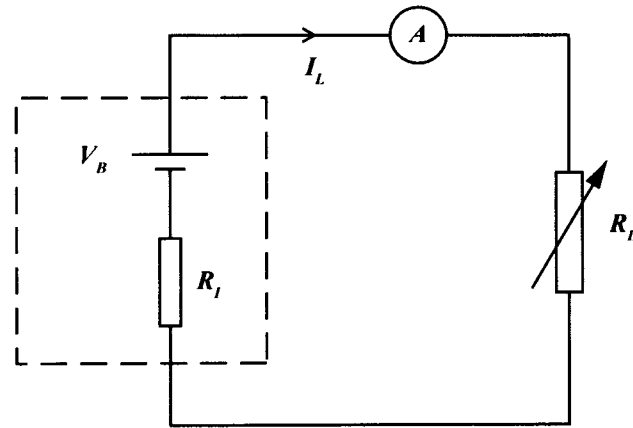


Figure 1. Circuit of resistance box

- (i) Measurements of current, I_L , are recorded as the external resistance, R_L , is varied. Show that for this circuit, the relationship between the measured current, I_L , and the external resistance, R_L , can be expressed by the equation

$$R_L = I_L^{-1} V_B - R_I.$$

[2 marks]



- (ii) The results of the experiment are shown in the first two columns of Table 1.

TABLE 1

R_L/Ohms	I_L/Amps	$(I_L)^{-1}/\text{Amp}^{-1}$
4.1	0.78	
5.9	0.67	
8.1	0.58	
10.3	0.51	
12.0	0.46	

Complete Table 1 by inserting the appropriate values in Column 3. [1 mark]

- (iii) On the grid provided in Figure 2 (page 9), plot a graph of I_L^{-1} versus R_L and draw the best straight line through the points. Choose appropriate scales to ensure that the origin is included. [4 marks]

- (iv) Determine and record the intercept on the I_L^{-1} axis.

[1 mark]

- (v) Use the intercept along with the equation given in (b) (i) to determine the internal resistance of the battery.

[2 marks]

Total 15 marks

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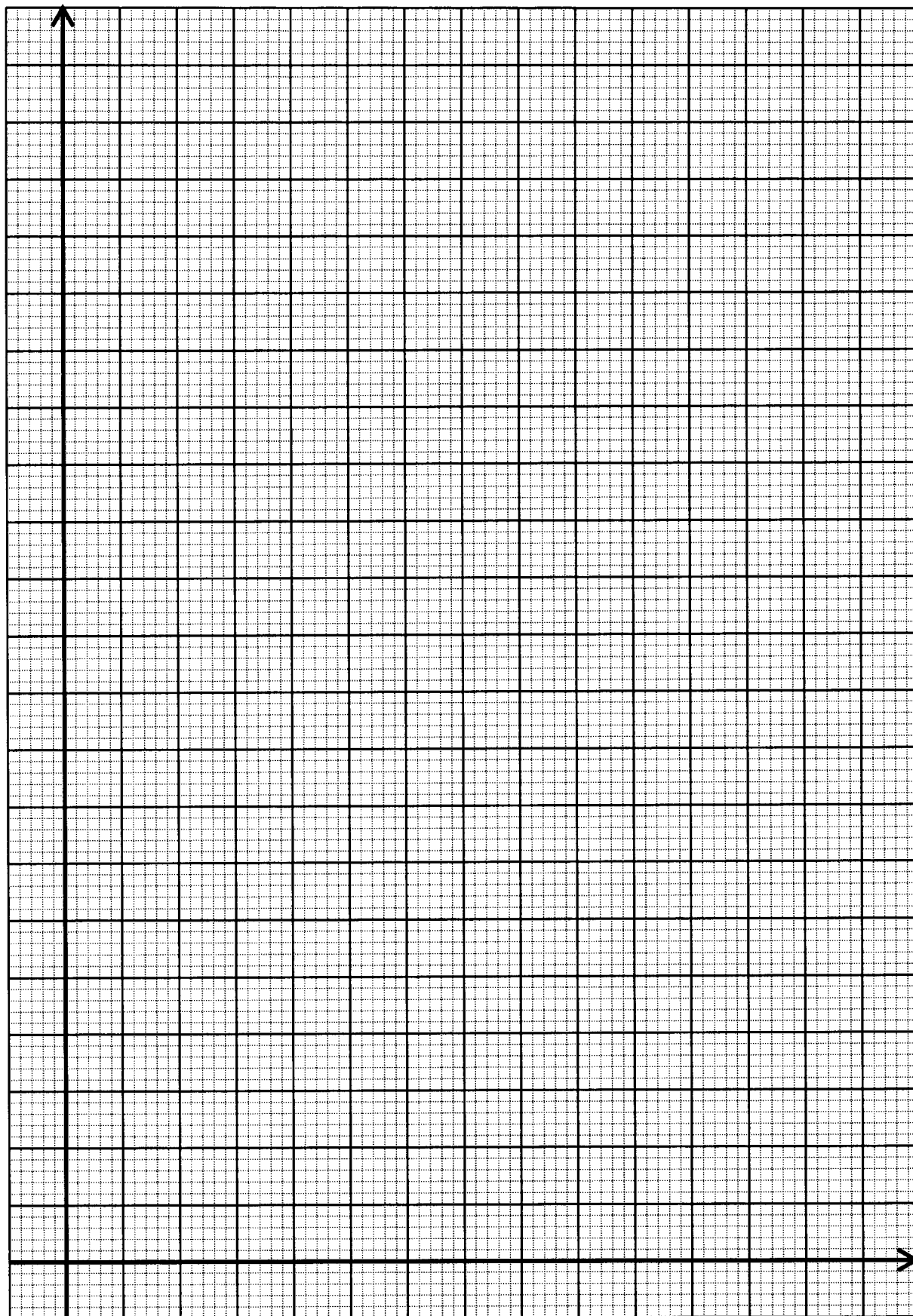


Figure 2. I_L^{-1} versus R_L

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2. (a) Draw the circuit diagram and derive the expression for the gain of a non-inverting amplifier.

[6 marks]



- (b) The circuit shown in Figure 3 is used in a tester for resistive transducers where the variation in resistance of the transducer is converted to a variation in voltage. R_x is a test resistor used to simulate the transducer.

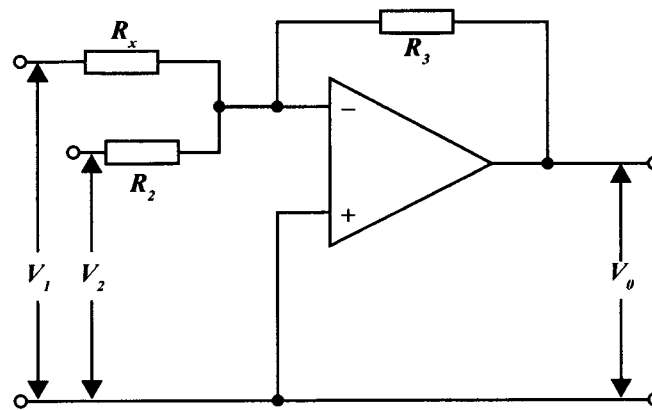


Figure 3. Circuit of tester

Show that output voltage V_o and R_x are related by the equation

$$V_o = 30\,000 R_x^{-1} - 10, \text{ where}$$

$$R_3 = R_2 = 10\text{ k}\Omega, \quad R_1 = R_x, \quad V_2 = 10\text{ V} \quad \text{and} \quad V_1 = -3\text{ V}.$$

[2 marks]



- (c) An experiment is carried out on the circuit shown in Figure 3 (page 11) in which the output voltage V_0 is monitored and recorded while R_x is varied over the range of values in Table 2.

TABLE 2

$R_x(k\Omega)$	$R_x^{-1}(10^{-3})$	V_0/V
2.0		6.0
3.0		1.0
4.0		-1.5
5.0		-3.0
6.0		-4.0

- (i) Complete Table 2 by inserting the appropriate values in Column 2. [1 mark]
- (ii) On the grid provided in Figure 4 (page 13), plot a graph of V_0 versus R_x^{-1} and draw the best straight line through the points. [4 marks]
- (d) Use your graph to determine the value of R_x which results in zero output voltage.

[2 marks]

Total 15 marks

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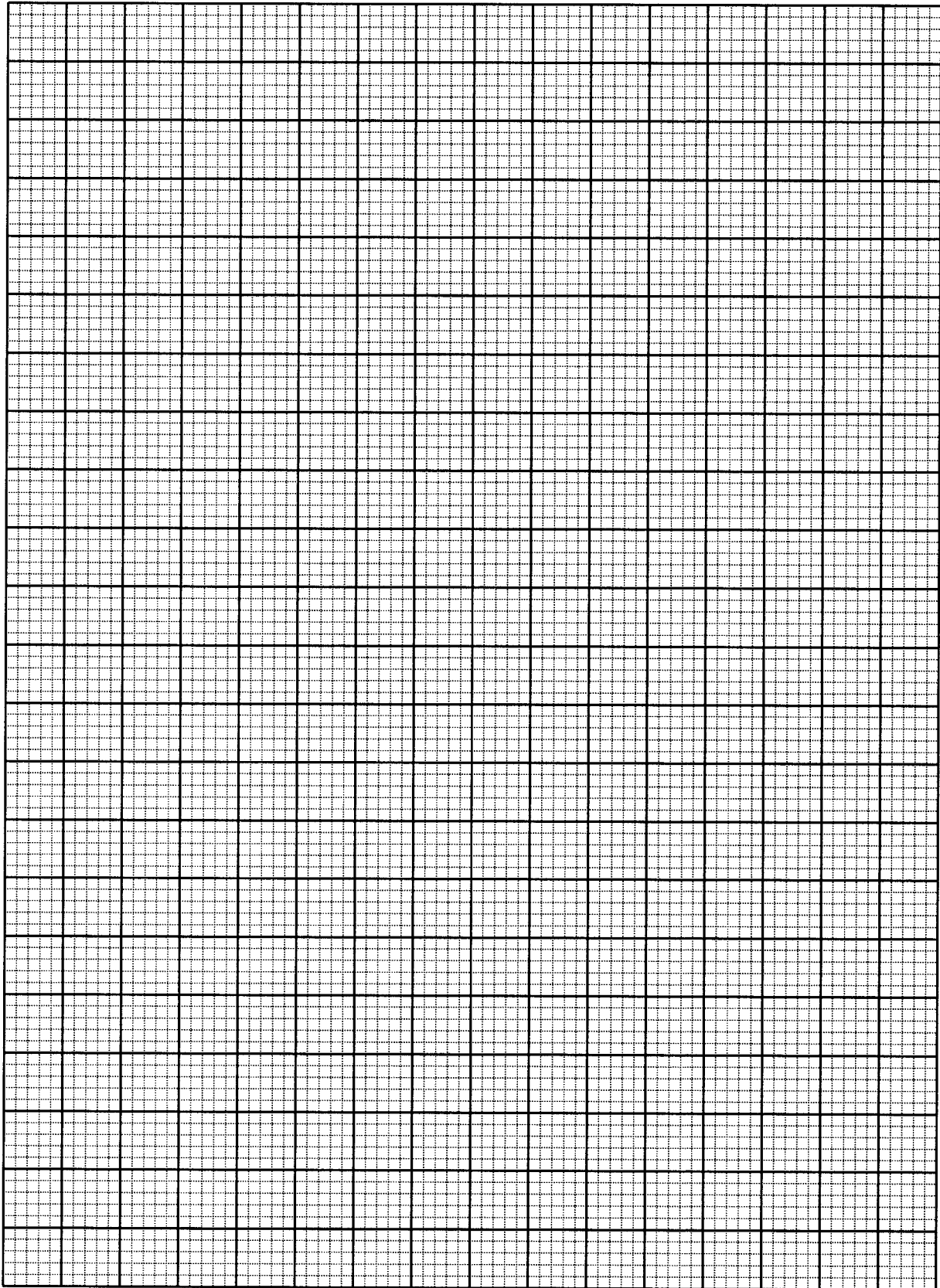


Figure 4. V_0 versus R_x^{-1}

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3. (a) X-rays are weakened by their passage through matter. Write the equation which quantifies this phenomenon, and state the meaning of EACH term in your equation.

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[4 marks]

- (b) Figure 5 is a schematic diagram of a setup which is used to measure the attenuation produced in a beam of x-rays by metal samples of different thickness.

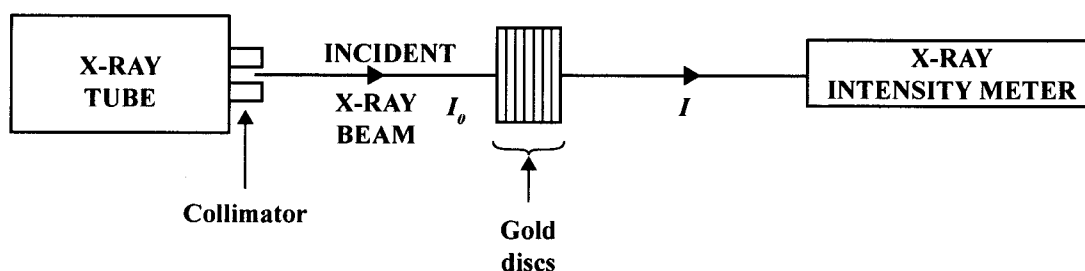


Figure 5. Setup to measure attenuation

In such a measurement, a beam of x-rays is directed at a target consisting of several thin gold discs of equal thickness of 0.4 mm. The ratio of emerging intensity, I , to incident intensity, I_0 , is recorded in Table 3 on page 16.

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TABLE 3

Attenuation (I/I_0)	Number of Discs	$\ln(I/I_0)^{-1}$	Total Thickness (mm)
1.00	0		
0.55	1		
0.30	2		
0.15	3		
0.09	4		
0.05	5		

- (i) Complete Table 3 by inserting the relevant values in Columns 3 and 4. [3 marks]
- (ii) On the grid provided in Figure 6, (page 17), plot a graph of the natural log of attenuation, $\ln(I/I_0)$ versus total thickness. [4 marks]
- (iii) Use your graph to determine the linear absorption coefficient of gold in cm^{-1} at the wavelength of the incident photons.

[3 marks]

- (iv) Suggest a reason why the linear absorption coefficient of a metal would decrease with decreasing wavelength of the incident beam.

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[1 mark]

Total 15 marks

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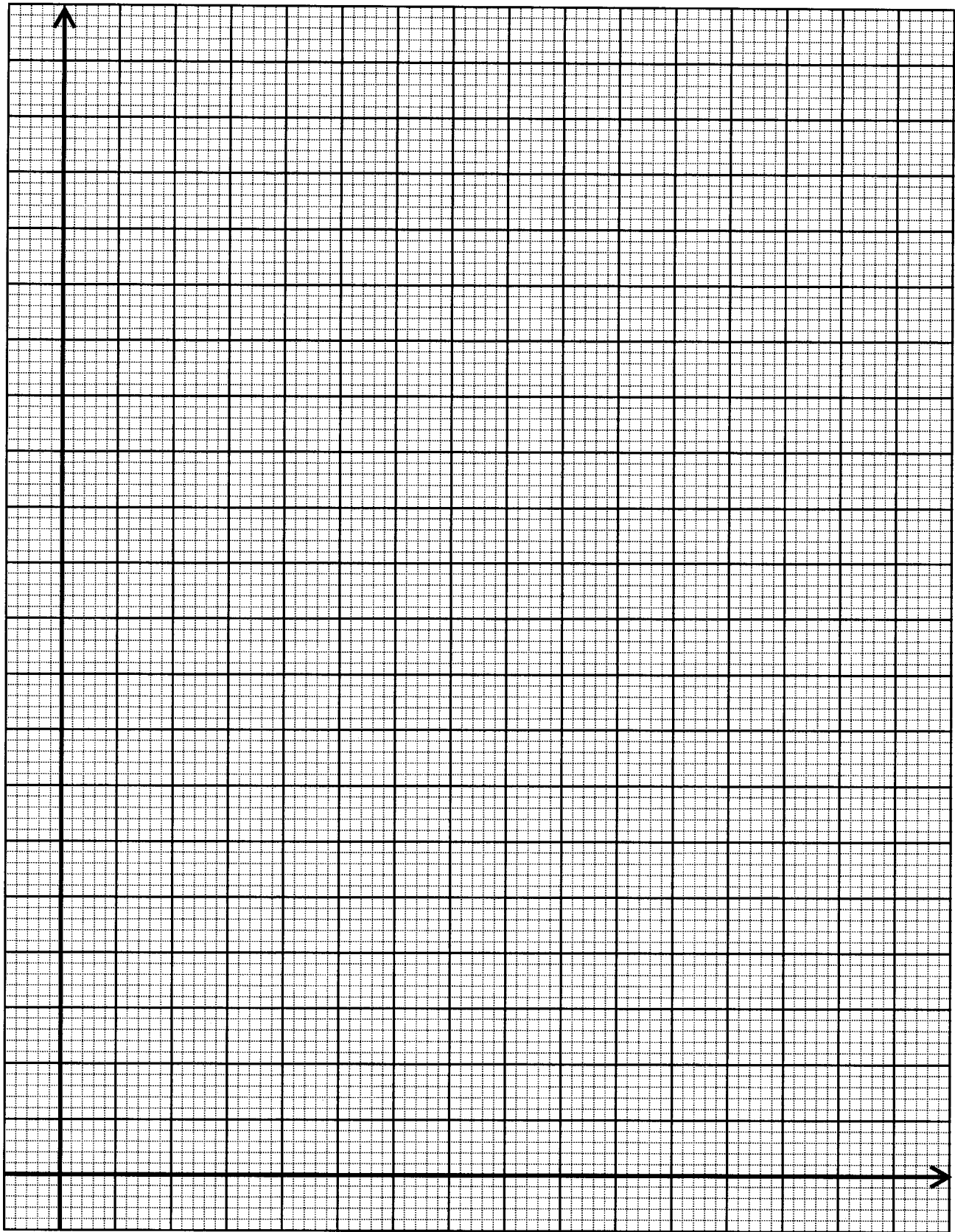


Figure 6. $\ln(I/I_0)$ versus total thickness

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SECTION B

Answer ALL questions.

Write your answers in the spaces provided.

4. (a) Explain the term 'drift velocity' when applied to the movement of electrons in a conductor.

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[2 marks]

- (b) Write the meaning of EACH variable in the following equation and state the unit in EACH case.

$$I = nevA.$$

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[4 marks]

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- (c) A cylindrical copper wire has the following specifications at ambient temperatures.

The wire carries a current of 6 A,

Radius = 6.9×10^{-4} m; Free electron density = $6.5 \times 10^{28} \text{ m}^{-3}$.

- (i) Calculate the drift velocity of electrons in the wire in **millimetres per second**.

[5 marks]

- (ii) Suggest a reason why this value is so small compared to the other values of electron velocity commonly encountered in physics.

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[2 marks]

- (iii) Would the drift velocity in a semiconductor material of similar dimensions carrying the same current be greater than that of the conductor in (c) (i)? State a reason for your answer.

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[2 marks]

Total 15 marks

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5. (a) (i) With the aid of a diagram, explain the operation of a simple half-wave rectifier circuit which employs a junction diode. Indicate the polarity of the output voltage on the diagram.

[6 marks]

- (ii) Sketch the output voltage when the input is a sine wave.

[1 mark]

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- (b) The voltage of the public electricity supply in some Caribbean countries can be described by the equation $V = 156 \sin 314 t$.

For such a supply voltage, determine

- (i) the peak voltage

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[1 mark]

- (ii) the frequency

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[2 marks]

- (iii) the root mean square (rms) voltage.

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[2 marks]

- (c) Draw a labelled sketch (do not plot) showing the correct values of amplitude and period for two cycles of the output voltage when this waveform is applied to a simple half-wave rectifier circuit.

[3 marks]

Total 15 marks

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- L

[8 marks]



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- (b) Cobalt-60 is an artificially produced isotope of Cobalt obtained by irradiating natural Cobalt-59 with fast neutrons from a nuclear reactor or similar source. Cobalt-60 is unstable and emits a beta particle and forms nickel. (Atomic numbers: Co – 27, Ni – 28)

Complete the following nuclear equations for this process.

Equation I: $\text{Co} + \text{n} \longrightarrow$

Equation II: $\longrightarrow$

[2 marks]

- (c) Cobalt-60 has a half-life of 5.27 years. Beginning with an initial sample of 200 grams of pure Co-60, determine how many grams of nickel would be found after three years.

Neglect the mass equivalence of the energy in the radiated decay products.

[5 marks]

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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EXTRA SPACE

If you use this extra page, you MUST write the question number clearly in the box provided.

Question No.

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